A real-time system

is one whose correctness is based on both the correctness of the outputs and their timeliness.

A system is a mapping of a set of inputs into a set of outputs.

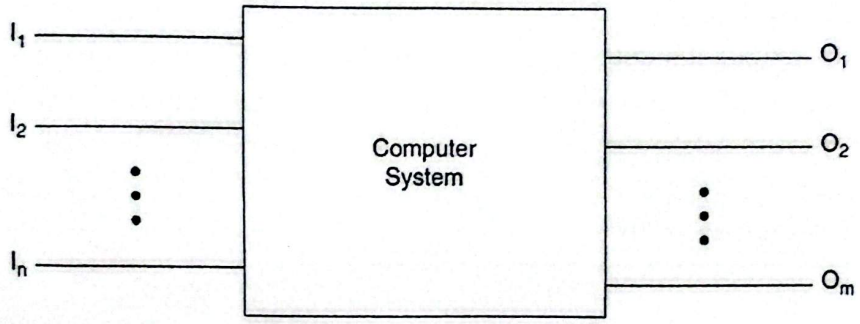


Figure 1.1 A system with *n* inputs and *m* outputs.

1- **Deadline of a job:** the time instant by which the task must complete execution.

Types of RTS:

1- **Hard real-time** — systems where the responses occur within the required deadline **(which means failure to meet a single deadline may lead to complete and catastrophic system failure).**

2- **Soft real-time** — systems where deadlines are important but which will still function correctly if deadlines are occasionally missed **(which means performance is degraded but not destroyed by failure to meet response-time constraints).**

3- **Firm real-time** — systems which are soft real-time but in which there is no benefit from late delivery of service **(which means a few missed deadlines will not lead to total failure, but missing more than a few may lead to complete and catastrophic system failure).**

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Table 1.1 A sampling of hard, soft, and firm real-time systems

	Real-Time Classification	Explanation
Automated teller machine	Soft	Missing even many deadlines will not lead to catastrophic failure, only degraded performance.
Embedded navigation controller for autonomous robot weed killer	Firm	Missing critical navigation deadlines causes the robot to veer hopelessly out of control and damage crops.
Avionics weapons delivery system in which pressing a button launches an air-to-air missile	Hard	Missing the deadline to launch the missile within a specified time after pressing the button can cause the target to be missed, which will result in catastrophe.

Real time system

Characteristics of RTS:

1-**Deterministic operations** are performed at predetermined times.

2- **User control** user specifies priority, what processes must always reside in main memory.

3- **Reliability** degradation of performance may have catastrophic consequences.

Questions

1- What are a system, real-time system, response-time of a system, and failure system?

2- What are the Characteristics of RTS?

3- What are the Types of RTS and compare between them?

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Lecture Two

Analog to Digital Converters

Analog to Digital Converters Analog to Digital Converters (ADC) are an electronic integrated circuit (IC) which transforms a signal from analog (continuous) to digital (discrete) form.

→Why ADC is needed:

because microprocessors can only perform complex processing on digitized signals.

Classification of ADC according to

1- The first group accuracy and speed: includes successive-approximation, counter, and flash-type converters.

2- The second group includes integrator converters and voltage to frequency converters

The tradeoff between two groups is based on accuracy VS speed.

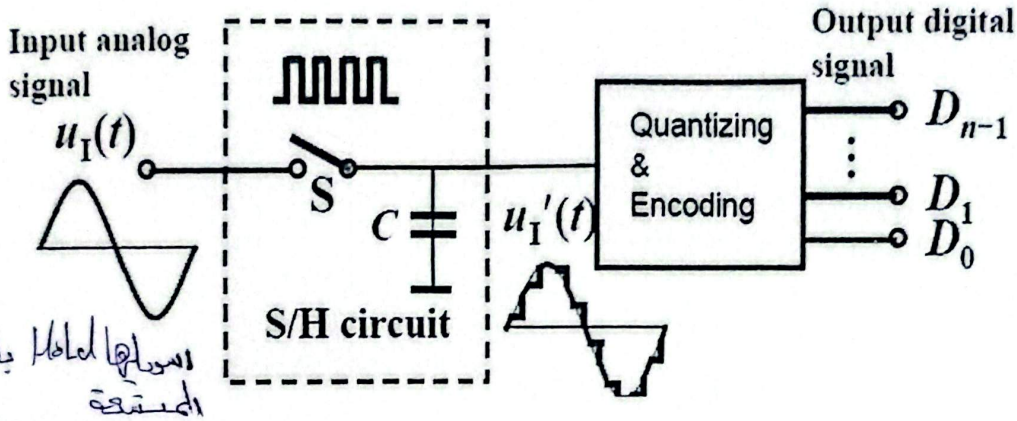
Note

successive-approximation and flash-type converters are faster but generally less accurate than the integrator and voltage to frequency type converters.

ADC process:

1. Sampling and Holding (S/H).
2. Quantizing and Encoding (Q/E).

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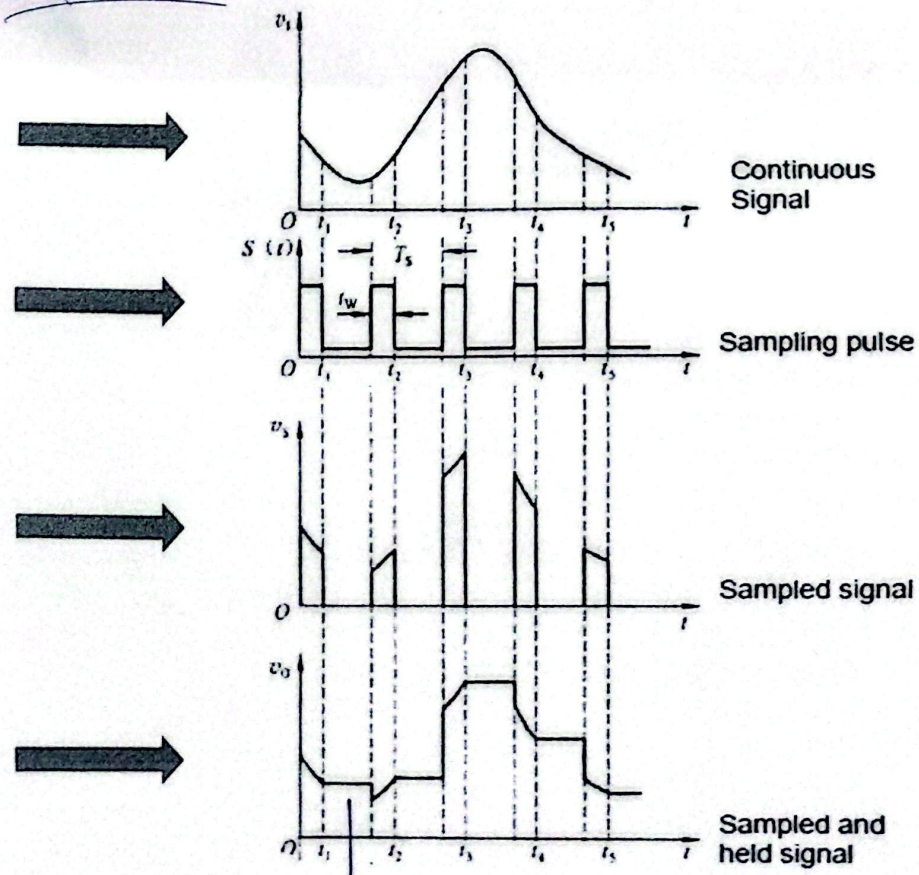
اسمها الحامل باستخدام المستقة

1- Sampling and Holding:

It converts analog signal into series of impulses, each representing amplitude of the signal at given point as shown in figure

Q// What Sampling and Hold?

الرسم مهم



خط مستقيم (ثابت) لان المسعة تحمل الحرة

n = عدد البتات
N = عدد الحالات الكلي تظهر

2- Quantizing and Encoding:

Quantizing - is the process of converting the sampled continuous signals into Discrete valued data (set of finite states).

Encoding - assigning a digital word or number to each state and matching it to the input signal.

The number of possible states that the converter can output is:

$N = 2n$ n is the number of bits in the ADC.

Example: if you have 0-10V signals. What is the Discrete Voltage Ranges and Output Binary Equivalent by using 3 bit A/D converter?

* اذا ما حدد الطريقة بالسؤال نكتفي بها بالطريقة

Solution:- $N = 2^n$ For a 3 bit A/D converter $N = 2^3 = 8$

quantization size: $Q = \frac{V_{max} - V_{min}}{N} = \frac{10\,V - 0\,V}{8} = 1.25\,V$

Step 1: Quantizing:

Step 2: Encoding

Output States	Discrete Voltage Ranges (V)	Output Binary
0	0 – 1.25 V	000
1	1.25 – 2.5	001
2	2.50 – 3.75 V	010
3		
4		
5		
6		
7	8.75-10.0	111

Ex: what the digital output if vin= 2.3

Ans:
Output= 001

نتوقف عند عدد البتات المذكور بالسؤال

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(n)

Resolution: Resolution is the number of bits used for conversion (8 bits, 12 bits, …,

Two ways to best improve accuracy of A/D conversion

1--Increasing the resolution

2--Increasing the sampling rate

Types of A/D Converters

1. Flash ADC
2. Delta-Sigma ADC
3. Dual Slope (integrating) ADC
4. Successive Approximation ADC

1. Flash A/D Converter

Fundamental Components

1- Resistors

use the resistors to form a ladder voltage divider, which divides the reference voltage into equal intervals.

2-Comparators

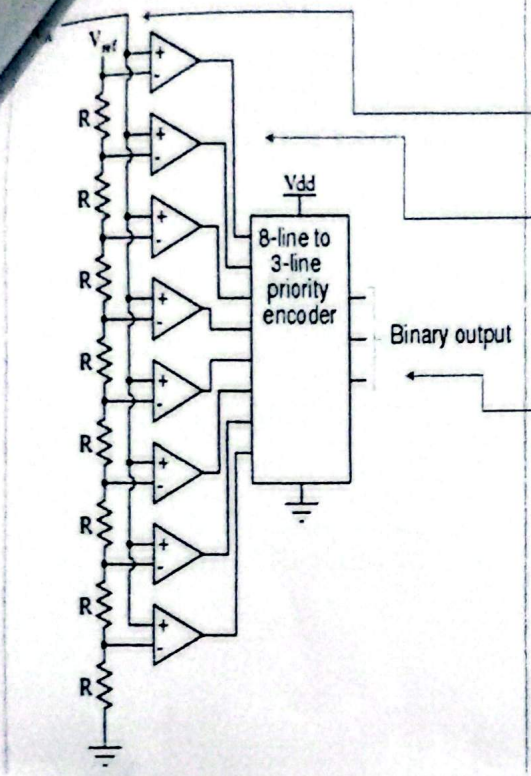
Consists of a series of comparators, which comparing the input signal to a unique reference voltage.

مشفر الاولويات

3-Priority encoder

the comparator outputs connect to the inputs of a priority encoder circuit, which produces a binary output.

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How does it work?

-- Uses the comparators to determine in which the input voltage Vin is exceed or not the Vref.

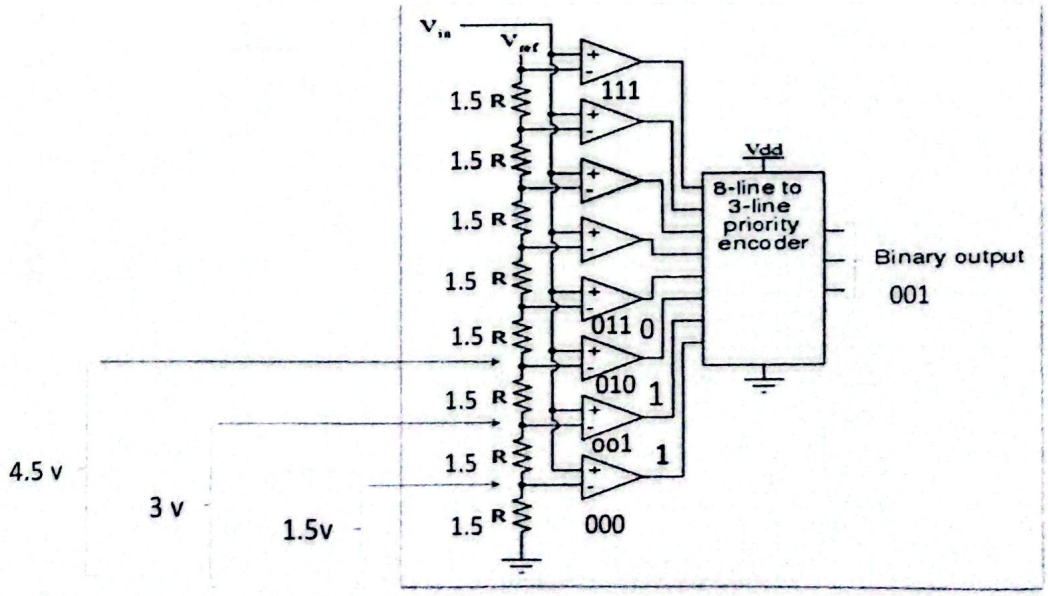
-- When the input voltage exceeds the reference voltage at each comparator, the comparator outputs will sequentially saturate to a high state.

--The priority encoder generates a binary number based on the highest-order active input, ignoring all other active inputs.

Q:Find the code-word for Vin=4.1V, and number of bits =3 with Vref=12V.

Solution: $N=2^3=8$ ⟹ $V_{ref}/N = 12/8 = 1.5\ V$

$If\ V_{in} > V_{ref}$ ⟹ **Comparator output = 1 else = 0**



8-

2. Successive Approximation ADC—

- A comparator and a DAC are used in the process.
- A Successive Approximation Register (SAR) is added to the circuit

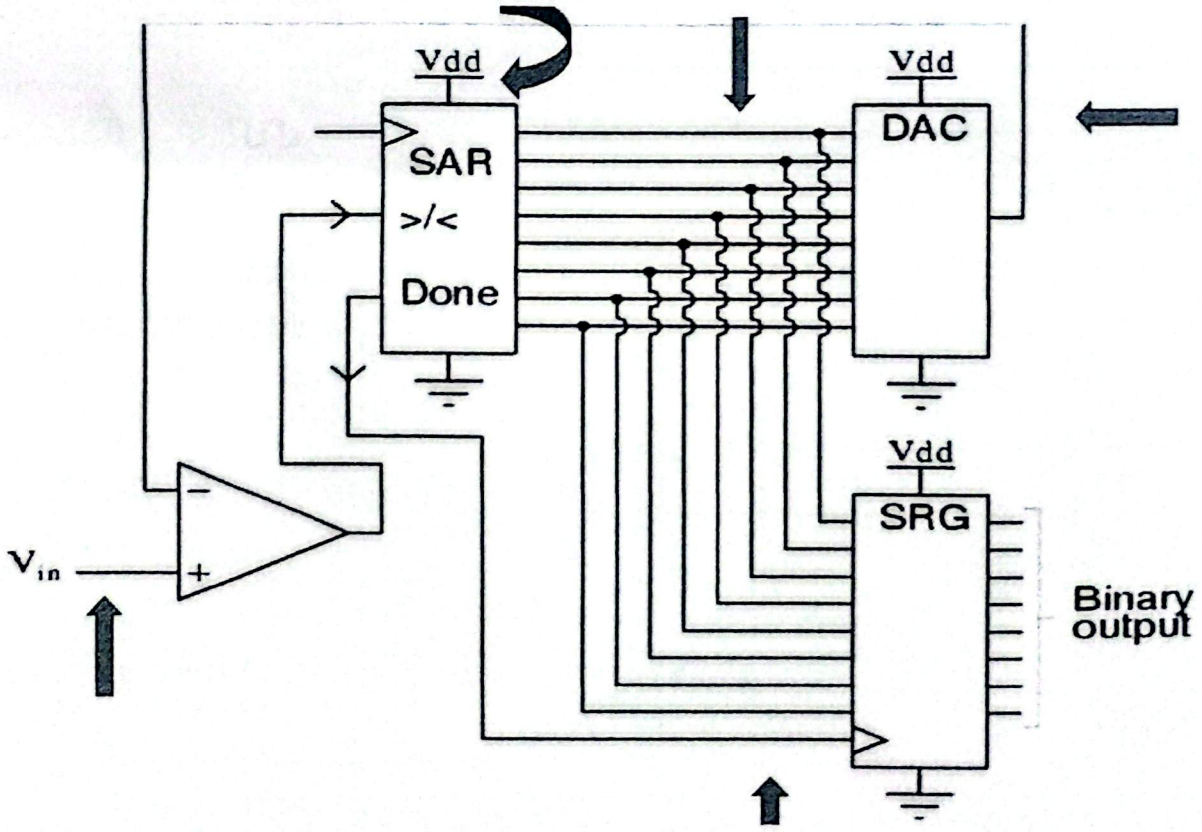
Components.

1- Comparator

2- SAR unit

3- DAC

4- Output Register.



Example 10 bit ADC, Vin= 0.6 volts (from analog device), Vref=1 volts, Find the digital value of Vin?

Solu: MSB (bit 9)

Divided Vref by 2

Compare Vref /2 with Vin

If Vin is greater than Vref /2 , turn MSB on (1)

If Vin is less than Vref /2 , turn MSB off (0)

Vin =0.6V & Vref /2= 0.5 V

1									

Since Vin>V, MSB = 1 (on)

Next Calculate MSB-1 (bit 8)

-- Compare Vin=0.6 V to V=Vref/2 + Vref/4= 0.5+0.25 =0.75V

-- Since 0.6<0.75, MSB is turned off

1	0								

Calculate MSB-2 (bit 7)

Go back to the last voltage that caused it to be turned on (Bit 9) and add it to Vref/8, and compare with Vin

Compare Vin with (0.5+Vref/8)=0.625

Since 0.6<0.625, MSB is turned off

1	0	0							

This process continues for all the remaining bits.

•**Digital Results:**

MSB	MSB-1	MSB-2	MSB-3	...					LSB
1	0	0	1	1	0	0	1	1	0

•**Results:** $\frac{1}{2}+\frac{1}{16}+\frac{1}{32}+\frac{1}{256}+\frac{1}{512}=.599609375$ V

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3--Sigma Delta ADC

.Main Components

1- Resistors

2- Integrator

3- Capacitor

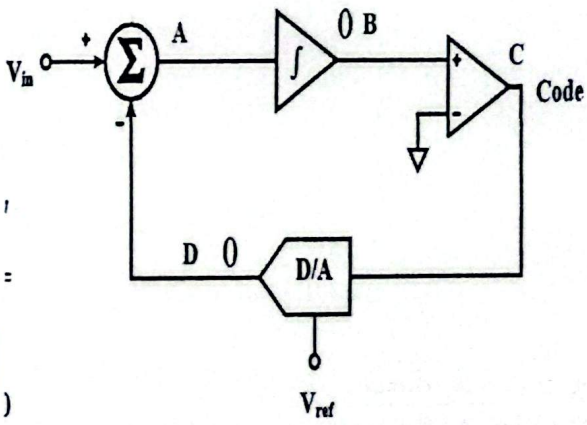
4- Comparators

5- Control Logic

EX

Find the code-word for a 5 bit, sigma-delta A/D with Vref=3.3V with Vin=2.7V

- $A_{new} = V_{in} - D_{old} = 2.7-0 = 2.7$
- $B_{new} = A_{new} + B_{old} = 2.7+0 = 2.7$
- If $B_{new}>0$, then C=1; else C=0
- If C=1, D=+V_{ref}; else D=-V_{ref}



Iteration 0	A	B	C	D
1	2.7	2.7	1	3.3
2	-0.6	2.1	1	3.3
3	-0.6	1.5	1	3.3
4	-0.6	0.9	1	3.3
5	-0.6	0.3	1	3.3

Bit Stream: 11111

Questions

1. What is an A/D Converter? What is the usual trade-off in A/Ds ?
2. What are the most common types? And how are they used?
3. What are the A/D processes? (draw or explain)
4. What is sample and hold?
5. What is Quantization and Encoding?
6. For a 4 bit A/D, with Vmin=1 V, V max=9 V, what is the code-word for Vin=3.1 Volts? What is the quantization error AVG, and quantization error Max.?
7. What is the resolution of an A/D?
8. How to increase the accuracy of an A/D?
9. How Does a Flash type A/D work?
10. What are the components of a Flash type A/D, draw the CCT.
11. Find the code-word for Vin=4.1V, and number of bits =3 with Vref=12V. (Use Flash type)
12. What are the components of a sigma-delta A/D, draw the CCT.
13. Find the code-word for a 5 bit, sigma-delta A/D with Vref=3.3V with Vin=2.7V.
14. Find the code-word for a 5 Bit SAR A/D with Vin=4.4V while Vref=12V.
15. What are the components of a SAR A/D, sigma-delta A/D, Dual-slope A/D and a flash-type A/D?

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